Ernest F. Aboagye, Ph.D., A.S.A.

Departments of Mathematics & Statistics
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Employment

Senior Lecturer, Actuarial Science, Departments of Mathematics & Statistics, Purdue University	2025–present

Industry Experience

Actuarial Analyst, Pension Consulting, Willis Towers Watson	2019–2020
Actuarial Analyst, Life Actuarial Consulting, Milliman	2018–2019
Actuarial Consultant, Pricing Analytics, New York Life Insurance Company	2017
Actuarial Intern, Commercial Lines Pricing, American Family Insurance	2017

Education and Credentials

Ph.D. in Risk Management & Insurance, Georgia State University – Maurice R. Greenberg School of Risk Science. Advisor: Prof. Liang Peng	2025
M.S. in Mathematics (Actuarial Science), Illinois State University	2017
B.S. in Actuarial Science & Statistics, University of Cape Coast, Ghana – Valedictorian	2014
Associate of the Society of Actuaries (A.S.A.), Society of Actuaries	2020

Research Interests

- Actuarial science, quantitative risk management, insurance economics, risk transfer
- Financial econometrics, data analytics and data science, AI in insurance and finance
- Enterprise risk management, emerging risk, catastrophe risk, climate risk, cyber risk

Publications and Manuscripts

Peer-reviewed Journal Articles

[1] **Aboagye, E.F.**, Asimit, V., Fung, T.C., Peng, L. and Wang, Q. (2025). A revisit of the optimal excess-of-loss contract. *European Journal of Operational Research*.

[2] **Aboagye, E.F.**, Ko, S.I.M., Lo, C.C., Hsiao, C.Y.L. and Peng, L. (2024). A contagion test with unspecified heteroscedastic errors. *Journal of Economic Dynamics and Control*.

Working Papers

[1] **Aboagye, E.F.** Is the past a reliable guide to the flood tail? Nonstationary event losses and the design of the NFIP's reinsurance layer. Submitted.

[2] **Aboagye, E.F.** Claim counts, claim sizes and excess-of-loss pricing under a common event shock. Submitted.

[3] **Aboagye, E.F.**, Koomson, J. and Boamah, E. More often or bigger? Exposure growth and tests for a shifting insurance loss tail. Submitted.

[4] **Aboagye, E.F.** and Yamoah, S.A. Volatility or contagion? Decomposing crisis changes in the marginal expected shortfall of insurers and banks. Submitted.

[5] **Aboagye, E.F.**, Anyomi, A.A. and Anyomi, S.K. Does index insurance fail when it matters most? Basis risk in the tail of U.S. area crop insurance. Submitted.

[6] **Aboagye, E.F.** Estimation risk in the optimal excess-of-loss contract. Working paper.

[7] **Aboagye, E.F.** and Peng, L. (2025). Model-based conditional risk forecasts. Working paper.

Work in Progress

[1] **Aboagye, E.F.** and Okine, A.N. More often, not bigger: The shifting tail of U.S. health data breaches and the third-party channel. Work in progress.

[2] **Aboagye, E.F.** and Huang, S. Does geographic pooling diversify flood risk? Spatial tail dependence and the tail index in U.S. flood insurance. Work in progress.

[3] **Aboagye, E.F.** Machine learning for the body, extreme value theory for the tail: Orthogonality and inference for conditional extreme quantiles. Work in progress.

Other Publications

[1] **Aboagye, E.F.**, Borketey, B. and Danquah, K. (2024). Optimizing travel insurance purchase detection using predictive models. *International Journal of Automation, Artificial Intelligence and Machine Learning*.

[2] **Aboagye, E.F.**, Borketey, B., Borketey, D. and Danquah, K. (2022). Predictive accuracy of the probability of credit card default using machine learning. *International Research Journal of Modernization in Engineering Technology and Science*.

Fellowships, Grants and Awards

Spencer Doctoral Fellowship, Spencer Educational Foundation	2024–2025
Center for the Economic Analysis of Risk (CEAR) Fellowship, Georgia State University	2022–2025
Huebner Foundation Scholarship Fellow, S.S. Huebner Foundation	2021–2025
James C. Hickman Scholar, Society of Actuaries	2021–2023
Helen C. Leith Foundation Scholarship, Georgia State University	2023–2025
Leyton B. Hunter Fellowship, Georgia State University	2023–2025
IABA Scholar, International Association of Black Actuaries	2016, 2021–2024
Greenberg School of Risk Science Research and Travel Grant, Georgia State University	2024
Graduate Fellowship & Assistantship, Georgia State University	2020–2025
Best Paper Award, Annual Statistical Project Competition, Illinois State University	2017
Valedictorian (Overall Best Graduating Student), University of Cape Coast	2014
Best Actuarial Student Award, Prudential Life Insurance Ghana	2014

Presentations and Conferences

Invited and Seminar Presentations

Purdue University, Departments of Mathematics & Statistics, *A revisit of the optimal excess-of-loss insurance contract*	2025
Bentley University, Department of Mathematical Sciences, *A revisit of the optimal excess-of-loss insurance contract*	2024
Illinois State University, Actuarial, Insurance & Statistics Seminar, *A revisit of the optimal excess-of-loss insurance contract*	2024
Georgia State University, Greenberg School of Risk Science, *A contagion test with unspecified heteroscedastic errors*	2023
Georgia State University, Greenberg School of Risk Science, *Optimizing travel insurance purchase detection using predictive models*	2022

Conference Participation

American Risk & Insurance Association (ARIA) Annual Meeting, Denver, US	2024
27th International Congress on Insurance: Mathematics & Economics (IME)	2024
American Risk & Insurance Association (ARIA) Annual Meeting, Online	2021
International Association of Black Actuaries Annual Conference	2016–2024

Other Talks

Speaker, Annual Celebration of the UCC Society of Actuarial Science Students: *The role of actuarial science in nation building*, Cape Coast, Ghana	2015

Teaching Experience

Instructor – Purdue University

STAT 472 – Fundamental Long-Term Actuarial Mathematics (SOA UEC)	Spring 2027
MA 373 – Financial Mathematics (SOA UEC)	Spring 2026, 2027
CS/STAT 242 – Introduction to Data Science	Spring 2026
MA/STAT 416 – Probability	Spring 2026
MA 158 – Precalculus	Fall 2025, 2026

Instructor – Georgia State University

AS 3230 – Financial Mathematics (SOA UEC)	Fall 2023
AS 4140L – Probability Lab	Fall 2022–Spring 2025
AS 3230L – Financial Mathematics Lab	Fall 2022–Spring 2025

Instructor and Teaching Assistant – Illinois State University

MAT 113 – Mathematical Reasoning Lab (Instructor)	Fall 2015, Spring 2016
MAT 280 – Financial Mathematics (TA)	Summer 2016
MAT 480 – Applications of Actuarial Principles (TA)	Fall 2016
MGT 100 – Statistical Reasoning (TA)	Fall 2015, 2016
MGT 227 – Operations Management (TA)	Spring 2016, 2017

Teaching Assistant – University of Cape Coast

MAT 101 – Algebra & Trigonometry	Fall 2014
MAT 203 – Further Calculus	Fall 2014
ACT 212 – Mathematics of Investment & Credit	Fall 2014
ACT 304 – Life Contingencies II	Spring 2015
ACT 401 – Survival Models	Spring 2015
ACT 404 – Pension Funds & Social Security	Spring 2015

Service

Peer Review

- Statistics: *Statistical Papers*
- Engineering: *IEEE-affiliated conference papers (EEEIC, AECE)*

Other Service

Quantitative Sciences Learning Community, Purdue University Indianapolis	2026
Judge, Stevie Awards for Women in Business	2024–2025
Recruitment Team, WTW Retirement Practice	2019–2020
Member, Technology Committee, International Association of Black Actuaries	2018–2020
Treasurer & Education Committee Chair, Actuarial Science Students Association, UCC	2012–2014

Professional Memberships

- Society of Actuaries (SOA)
- American Risk & Insurance Association (ARIA)
- Risk and Insurance Management Society (RIMS)
- American Finance Association (AFA)
- International Association of Black Actuaries (IABA)

Skills

- Software: R, Python, SAS, Excel VBA, MATLAB, Stata, LaTeX; actuarial platforms (Quantify, Prophet, MG-ALFA)
- Languages: English (fluent); Twi, Ga and Fante (native)
